

SARATH KUMAR SENTHILKUMAR KAVITHA

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PROFILE

MSc Robotics student at the University of Manchester and a first-class BTech AI and Robotics graduate, bringing a genuine passion for building systems that work in the real world. Experienced in programming autonomous robots with ROS and ROS2, deploying machine learning models on embedded hardware, and integrating sensor systems for real-time perception and control. Three internships across robotics engineering, autonomous navigation and process automation have given me solid practical foundations alongside my academic work.

PROFESSIONAL EXPERIENCE

Robotics Engineer Intern **Bimetal Pvt. Ltd. (VIT Contractual Project)** Chennai, India *Jan 2025 to Aug 2025*

- ▶ Built a real-time surface defect detection system from the ground up, running TensorFlow and YOLO inference on an NVIDIA Jetson Xavier embedded computer. Achieved 85% detection accuracy in an industrial production environment with no cloud connectivity.
- ▶ Redesigned the preprocessing pipeline and CNN feature extraction approach, lifting detection precision by 89%.
- ▶ Integrated the AI system with a robotic spray mechanism that automatically marked defective areas, creating a closed-loop, edge-to-actuation pipeline. Collaborated with three engineering teams to bring it into their existing inspection workflow.

ROS2 Programming Intern **Mafkin Robotics** Chennai, India *Jun 2024 to Aug 2024*

- ▶ Programmed the localization and motion control systems for an autonomous ship-cleaning robot using ROS2. Fused data from a camera, IMU and wheel encoders to achieve real-time mapping, obstacle avoidance and smooth path following.
- ▶ Designed and tuned control algorithms to hit the precision localization targets the project required, building the kind of sensor integration experience that transfers directly to autonomous vehicle and ADAS development.

RPA Developer Intern **Tata Consultancy Services** Chennai, India *Aug 2023 to Nov 2023*

- ▶ Designed and deployed 15 automated business process workflows using UiPath, reducing manual workload significantly. Earned the UiPath Certified RPA Developer credential during the placement.

PROJECTS

Leo Rover Autonomous Navigation (MSc Robotics, University of Manchester) *April 2026*

- ▶ Built a Leo Rover system integrating a robotic arm, Raspberry Pi, depth camera and 2D-LiDAR for fully autonomous navigation and manipulation. Implemented ROS2-based frontier exploration for autonomous environment mapping, achieving localization accuracy within ± 12 cm on hardware with real-time path planning and obstacle avoidance.
- ▶ Developed a computer vision pipeline using the depth camera to detect colored blocks in real time, with manipulator control logic to autonomously pick up blocks and place them in designated positions. Integrated sensor fusion across LiDAR, depth camera and IMU for reliable perception indoors, validated in Gazebo before hardware deployment.

Cascaded PID Drone Control *March 2025*

- ▶ Designed and implemented a cascaded PID control system for autonomous drone stabilization, with an inner loop regulating angular rate and an outer loop controlling attitude and altitude. Tuned control gains through iterative simulation testing before successfully deploying and validating the controller on a real drone in live flight tests.
- ▶ Validated the full quadrotor dynamics model in Python simulation, then transferred the controller to hardware, achieving stable hover and responsive setpoint tracking on the physical drone. Demonstrated robust recovery from external perturbations in real-time flight, confirming the sim-to-real transfer of the control design.

Robotic Automation for Fault Detection in Metal Sheets

Aug 2025

- ▶ Designed and validated a complete edge AI pipeline: OpenCV preprocessing, YOLO object detection and TensorFlow classification running on NVIDIA Jetson Xavier, achieving 89% defect identification accuracy and improving throughput by 40%.
- ▶ Connected the inference output to a physical robotic spray actuator, demonstrating closed-loop embedded AI system design from sensor to actuation.

Nuclear Environment Automation System

Dec 2024

- ▶ Designed an autonomous robot for operation in hazardous nuclear environments, using Visual SLAM for real-time mapping and localisation. Validated navigation accuracy and sensor performance through thorough Gazebo simulation testing before any hardware work.
- ▶ Embedded depth cameras for perception and attached a robotic arm for autonomous task execution, combining manipulation and navigation in a single system.

Agri-Connect Smart Irrigation System

Aug 2023

- ▶ Built a Raspberry Pi IoT system integrating soil moisture, rainfall, humidity and temperature sensors with real-time automation algorithms. Used MQTT for device communication and exposed a web interface for remote monitoring and manual override.

EDUCATION

MSc. Robotics

The University of Manchester Manchester, UK

Sep 2025 to Present

Modules include: Robotic Systems, Computer Vision, Reinforcement Learning, Autonomous Mobile Robots, Robotic Manipulators, Cognitive Robotics, Software for Robotics

BTech. Computer Science Engineering with AI and Robotics

Vellore Institute of Technology Chennai, India

Sep 2021 to Jul 2025

CGPA: 9.07 / 10.0 (First-Class Honours) Modules include: Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing, Sensors and Actuators, Robotics Kinematics, Robot Operating System (ROS), Cognitive Robotics, Robotics Perception

TECHNICAL SKILLS

AI and ML: TensorFlow, YOLO, Deep Learning, CNN, OpenCV, Computer Vision, NLP, Reinforcement Learning

Languages: Python, C++, C, JavaScript, Java, SQL

Robotics: ROS1, ROS2, Gazebo, RViz, SLAM, Sensor Fusion, Nav2, MoveIt2

Embedded Systems: NVIDIA Jetson Xavier, Raspberry Pi, Arduino, Real-Time Systems, IoT

Protocols: MQTT, HTTP, I2C, UART

Frameworks: ReactJS, PyQt5, Firebase, UiPath RPA

AWARDS AND CERTIFICATIONS

- ▶ 1st Place at CSI One-Day Product Development Hackathon 2023, Computer Society of India
- ▶ Stage 3 Finalist at e-Yantra Robotics Competition 2024, IIT Bombay, placing in the top 100 teams globally
- ▶ UiPath Certified RPA Developer Foundation, October 2023
- ▶ Google Cloud Computing Foundation (NPTEL), November 2023
- ▶ Enterprise Automation Certificate, Workato, June 2023